

Data preprocessing report: Loan prediction analysis

1. Introduction

Data preprocessing is an important stage in preparing a dataset for machine-learning analysis. Raw datasets may contain missing values, irrelevant variables, categorical information, unusual observations, and numerical variables measured on different scales. If these issues are not addressed appropriately, they can affect the quality, reliability, and interpretability of a machine-learning model. For this project, the Loan Prediction dataset was prepared for a binary classification task in which the aim is to predict whether a loan application is approved or rejected. The preprocessing process included checking the quality of the data, treating missing values, checking duplicates and data types, removing the irrelevant Loan_ID variable, encoding categorical variables, detecting potential outliers, splitting the data into training and testing sets, scaling numerical features, and selecting relevant features. The decisions made during preprocessing were based on the characteristics of the dataset and the requirements of the modelling task. The objective was not simply to modify the data, but to produce a reliable dataset while preserving information that could be useful for predicting loan approval.

2. Missing value treatment

The dataset was first inspected for missing values because incomplete observations can cause problems during machine-learning modelling. Missing values were identified in variables including Gender, Married, Dependents, Self_Employed, LoanAmount, Loan_Amount_Term, and Credit_History. The missing values were treated during preprocessing so that the final modelling data would not contain unresolved missing observations. This was important because many machine-learning algorithms require complete numerical input. Treating missing values rather than removing every incomplete record also helps preserve the available sample size and reduces the possibility of losing potentially useful applicant information. After preprocessing, a final check confirmed that the feature dataset contained zero missing values. This confirmed that missing value treatment had been completed successfully.

3. Duplicate and data type checks

Duplicate records were checked to determine whether the dataset contained repeated applications that could distort the analysis or cause some observations to receive greater

influence during modelling. The dataset was also inspected to confirm that variables were stored in suitable data types. These checks were necessary because data quality problems can affect model performance even when the values themselves appear reasonable. Confirming the structure and uniqueness of the observations provided a reliable foundation for the remaining preprocessing steps.

4. Removal of irrelevant features

The `Loan_ID` variable was removed from the modelling dataset. Although it identifies individual loan applications, it does not describe an applicant's financial position, loan characteristics, or creditworthiness. Keeping an identification number could introduce meaningless patterns into a machine-learning model and would not provide useful information for predicting `Loan_Status`. Removing `Loan_ID` therefore reduced unnecessary information and ensured that the model focused on variables with potential predictive value.

5. Categorical variable encoding

Several variables in the dataset were categorical, including `Gender`, `Married`, `Dependents`, `Education`, `Self_Employed`, and `Property_Area`. Machine-learning algorithms require these categories to be represented numerically, so the categorical variables were transformed using one-hot encoding. One-hot encoding was appropriate because it represents each category as a separate binary feature without suggesting that one category is numerically greater or smaller than another. For example, `Property_Area` was transformed into features such as `Property_Area_Semiurban` and `Property_Area_Urban`. This feature engineering step converted the original categorical information into a format that could be used effectively by the machine-learning algorithms.

6. Target variable encoding

The target variable, `Loan_Status`, originally contained the categories `Y` and `N`. It was converted into a binary numerical variable, where 1 represents an approved loan and 0 represents a rejected loan. This transformation was necessary because the project is a binary classification problem. A binary numerical target allows the machine-learning model to distinguish clearly between the two possible outcomes while preserving the original meaning of the variable. The resulting target distribution contained 422 approved applications and 192 rejected applications.

7. Outlier detection and handling

Potential outliers were investigated in the numerical variables using boxplots and the Interquartile Range (IQR) method. The IQR method identifies observations that fall below $Q1 - 1.5 \times IQR$ (lower_bound) or above $Q3 + 1.5 \times IQR$ (upper_bound). It was selected because it is a practical method for identifying unusual observations without requiring the variables to follow a normal distribution. Potential outliers were identified in ApplicantIncome, CoapplicantIncome, and LoanAmount. Loan_Amount_Term also contained values that differed from the dominant 360-month repayment period. The potential outliers were not automatically removed. This decision was important because an unusual value is not necessarily an incorrect value. For example, an applicant with a particularly high income or loan amount may be a genuine applicant rather than a data entry error. Removing such observations without sufficient evidence could reduce the representativeness of the dataset and remove useful information. The observations were therefore retained while being recognised as potentially influential values.

8. Train-test split

The processed dataset was divided into training and testing subsets. The final split produced 491 observations for training and 123 observations for testing. The purpose of this split was to ensure that the model could be trained on one portion of the data and evaluated on observations that were not used during training. This provides a more realistic indication of how the model may perform on new loan applications. The target distributions were also checked after splitting. The training set contained 342 approved and 149 rejected applications, while the testing set contained 80 approved and 43 rejected applications. This confirmed that both target classes were represented in each subset.

9. Feature scaling

Feature scaling was applied to the numerical variables ApplicantIncome, CoapplicantIncome, LoanAmount, and Loan_Amount_Term. These variables have different numerical ranges, and scaling helps place them on a comparable basis. StandardScaler was selected because it standardizes numerical variables using their mean and standard deviation. This was considered appropriate because the numerical variables are measured on different scales and some contain potential extreme observations. The scaler was fitted using the training data and then used to transform the relevant data. This approach is important because fitting the scaler using the complete dataset could allow information from the testing data to influence the preprocessing process, resulting in data leakage.

10. Feature selection and correlation analysis

Feature selection was supported by a correlation heatmap and model-based feature importance analysis. The correlation heatmap was used to examine relationships between numerical variables and identify variables that might contain highly overlapping information. The strongest observed relationship was between ApplicantIncome and LoanAmount, with a correlation coefficient of approximately 0.57. This represents a moderate positive relationship, suggesting that applicants with higher incomes tended to have higher loan amounts. However, the relationship was not considered strong enough to justify removing either variable because both may contain useful information for predicting loan approval. The remaining numerical variables generally showed weaker relationships. Therefore, no predictor was removed solely because of correlation.

11. Feature importance analysis

A Random Forest classifier was used to evaluate the relative importance of the available features. This method was selected because it can capture nonlinear relationships and interactions between variables and provides a model-based measure of feature importance. The analysis identified Credit_History as the most important feature, followed by ApplicantIncome, LoanAmount, CoapplicantIncome, and Loan_Amount_Term. This indicates that credit history and financial characteristics contain substantial predictive information for the loan approval outcome. The feature importance results were used as supporting evidence rather than as the sole basis for removing variables. Features with lower importance were retained because low importance in one model does not necessarily mean that a variable contains no useful information.

12. Key observations

The preprocessing and feature selection stages produced several important observations. Credit_History was the strongest feature according to the Random Forest analysis, highlighting the importance of an applicant's credit history in predicting loan approval. ApplicantIncome and LoanAmount were also among the most influential features, indicating that financial characteristics are important to the prediction task. The correlation analysis showed a moderate relationship between ApplicantIncome and LoanAmount but did not identify a sufficiently strong relationship to require feature removal. Potential outliers were found in several numerical variables, but they were retained because they could represent legitimate applicant characteristics. The encoding process successfully converted

categorical information into machine readable features. After preprocessing, the final feature dataset contained no missing values and 14 predictor variables.

13. Final machine-learning dataset

The final machine-learning dataset contains 614 observations and 14 predictor features. The predictor variables are ApplicantIncome, CoapplicantIncome, LoanAmount, Loan_Amount_Term, Credit_History, Gender_Male, Married_Yes, Dependents_1, Dependents_2, Dependents_3+, Education_Not_Graduate, Self_Employed_Yes, Property_Area_Semiurban, and Property_Area_Urban. Loan_Status was retained separately as the target variable and encoded as 1 for approved applications and 0 for rejected applications. The final feature dataset contains zero missing values. The final dataset is therefore structured, encoded, checked, and prepared for the machine-learning modelling stage. The preprocessing decisions also preserve the original applicant information as far as possible while removing irrelevant information and addressing issues that could affect modelling.

14. Conclusion

The preprocessing stage transformed the original Loan Prediction dataset into a machine-learning ready dataset through a series of data quality, feature engineering, outlier, scaling, and feature selection steps. Each method was selected based on the nature of the data and the requirements of the classification problem. The final dataset contains 614 observations, 14 predictor features, an encoded target variable, and no missing values. The analysis also identified Credit_History, ApplicantIncome, LoanAmount, CoapplicantIncome, and Loan_Amount_Term as important predictive features. Overall, the preprocessing process provides a suitable foundation for developing and evaluating machine-learning models for loan approval prediction.